Siavash (Isaac) Barqi Janiar

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Experience

Sr. Data Analyst | Rogers Communications Sep. 2025 – Present

- Developed **PySpark and Python** data pipelines and deployed on **Azure Cloud** that **accelerated large-scale data processing by 73.5%**, **reducing compute time by 2.4x**.
- Leveraged **Azure OpenAI** to develop AI-powered analytics solutions and built **Scikit-learn forecasting models** to predict operational trends, improving forecast **accuracy by 90%** and **reducing manual analysis time by 60% (3 days per week)**.
- Developed complex **SQL queries for MySQL** on **Databricks** to process +10M-row datasets, cutting data retrieval time from minutes to seconds through **indexing and query optimization**.
- Designed and maintained **Power BI** dashboards used by **200+ stakeholders**, improving **reporting efficiency by 35%** through optimized **DAX measures** and **Power Query** data modeling.
- Built automated ETL workflows in **Python and Power Automate**, reducing manual data preparation time by **50–70% per report cycle**.
- Led cross-functional analytics initiatives, delivering insights that influenced decisions resulting in **+$50K in cost savings in 2026**.
- Partnered with business leaders to create **KPI and SLA** frameworks, leading to a **20% improvement** in performance tracking visibility across departments.

Data Engineer | Docma Jan. 2024 – Apr. 2025

- Designed and deployed full-scale **ETL pipelines** from ground up using **advanced Python coding**, including a **custom CLI tool**, **data preprocessing modules**, an **alpha-lag-decay forecasting model (implemented in R)**, and **export functionality**, resulting in **84.1% sales attribution prediction accuracy**.
- **Designed and optimized Snowflake and PostgreSQL database architectures**, achieving **2.55× faster query execution** and a **25% reduction in compute resource usage** through **strategic indexing, table restructuring, and query optimization**.
- **Led client-facing meetings** to deliver product demos, gather requirements, and communicate roadmap updates; ensuring 100% client feedback incorporation into engineering priorities and fostering **stronger stakeholder relationships**.
- Utilized **Power BI** to design and develop insightful **data visualizations and executive-level slide decks**, effectively communicating key trends and analytical findings to **senior data scientists and high-level decision makers** for strategic planning and data-driven decision-making.
- **Automated 80% of recurring data workflows**, reducing manual effort by **12+ hours per week** enabling **self-service analytics** across teams and significantly accelerating reporting and analysis cycles, increasing team productivity and consistency across environments.
- Integrated data ingestion and export across five sources, **local files**, **integrated APIs**, **AWS S3**, **Google Cloud Storage**, and **PostgreSQL databases**; enabling seamless, multi-source data synchronization and improving pipeline flexibility.
- Implemented comprehensive **documentation and code structure** using **Markdown** and **PEP 257-compliant docstrings**, integrated with version-controlled CI/CD pipelines via **Git to improve maintainability**. **Reduced manual setup time by 70% and increased team productivity and consistency across development and production environments**.

Research Assistant | York University Sep. 2021 – Feb. 2024

- Proposed a **transfer learning (TL)** method based on **feature extraction** to predict jamming patterns in a communication network. Reduced the time complexity of the primary model by **30 times**.
- Realized a comprehensive **eXplainable AI (XAI)** method comprising **pattern recognition** and **rule learning** for network security. Improved the **transparency** of the model compared to the benchmark explainable models **by 17%**, whilst having a **32% less error rate**.

- Introduced an evaluation environment comparing the performance of **Recurrent Neural Network (RNN) structures** utilizing **Long Short-Term Memory (LSTM)** layers, achieving a **13% higher throughput rate** than **Convolutional Neural Network (CNN) structures** while having **x1.2 fewer parameters**.
- Implemented a **Transformer-based model** combined with **K-means clustering** to handle sparse data, using filtering and **PCA** as preprocessing steps, and achieved **88% accuracy** in annotating correct jamming pattern groups.

Research Assistant | Amirkabir University Feb. 2020 – Jan. 2021

- Realized an efficient **model-free reinforcement learning** MAC protocol for frequency resource allocation. Surpassed the benchmark protocol with nearly **60% better throughput**.
- Leveraged an **online actor-critic** algorithm for access problems in heterogeneous networks. Achieved **95% throughput** in the network marked as the **highest possible performance**.
- Optimized the resource allocation system in distributed computer networks with prioritized packets using ML/AI, which increased the throughput of the wireless system **by approximately 15%**.

Skills

Programming Languages: Python, SQL, C/C++, Java, R, MATLAB, JavaScript, Node.js, C#
Frameworks: Power BI, Snowflake, Data Bricks, AWS, GCP, Pandas, Docker, Sci-kit learn, PostgreSQL, Dash, MatPlotLib, Plotly, Psycopg2, AWS Wrangler, Apache Kafka, Spark, Regex, Boto3, PyTorch, Flask, Postman, Git

Selected Projects

Real-Time Stock Market Data Platform (Golden Rod) — Python, C++, SQL, Dash, Vue.js [Link]

- Implemented a **real-time data analysis pipeline** to ingest, transform, and load stock exchange data using **advanced Python and thread programming**, supporting continuous updates and ensuring data consistency across components.
- Used **Apache Kafka** to build a scalable data streaming architecture, improving system reliability and enabling efficient processing of real-time stock market data.
- Optimized **PostgreSQL** for analytical workloads and containerized the environment using **Docker**, leveraging indexed time-series partitions, materialized views, and query plan tuning to reduce data retrieval time.
- Developed a **RESTful API** using **Flask** and **Vue.js** to deliver real-time stock market predictions. Integrated **Apache ECharts** to enable users to interact with **over 10 types of dynamic charts and data visualizations**.

RAG-based Chatbot — LangChain / FAISS [Link]

- Built a **RAG-based chatbot** leveraging **LangChain** and **OpenAI language models** to provide context-aware, retrieval-augmented responses from custom knowledge sources.
- Implemented semantic search using a **FAISS retriever** with a **lightweight embedding model** for efficient vector-based document retrieval.
- Preprocessed documents using **recursive text splitter** for optimal chunking and embedded them into a **Chroma vectorstore** for fast and scalable retrieval.
- Deployed the chatbot using **Streamlit**, delivering an interactive and production-ready interface for end users.

Recent Publications [Link]

- S. B. Janiar, P. Wang, "Intelligent Anti-jamming based on Deep Reinforcement Learning and Transfer Learning," *IEEE Transactions on Vehicular Technology*, 2023.
- S. B. Janiar, Xian Lu, P. Wang, "Explainable Reinforcement Learning for Wireless Security at the Physical Layer: A Survey," IEEE Transactions on Wireless Communications, 2022.
- Barqi Janiar S, Pourahmadi V, "Deep-reinforcement learning for fair distributed dynamic spectrum access in priority buffered heterogeneous wireless networks," *IET Commun. 2021;19*. https://doi.org/10.1049/cmu2.12098
- S. B. Janiar, P. Wang, "A transfer learning approach based on integrated feature extractor for anti-jamming in wireless networks," *IEEE PIMRC, Toronto*, 2023.

Education

York University | Master of Science in Computer Science Sep. 2021 - May 2023
Field of Study: AI and Machine Learning GPA: 8.8/9
Amirkabir University | Bachelor of Science in Electrical Engineering Sep. 2016 - Jun. 2021
Focus: AI and Machine Learning GPA: 3.74/4